

Q12. Scenario: Monthly Temperature and Rainfall Data

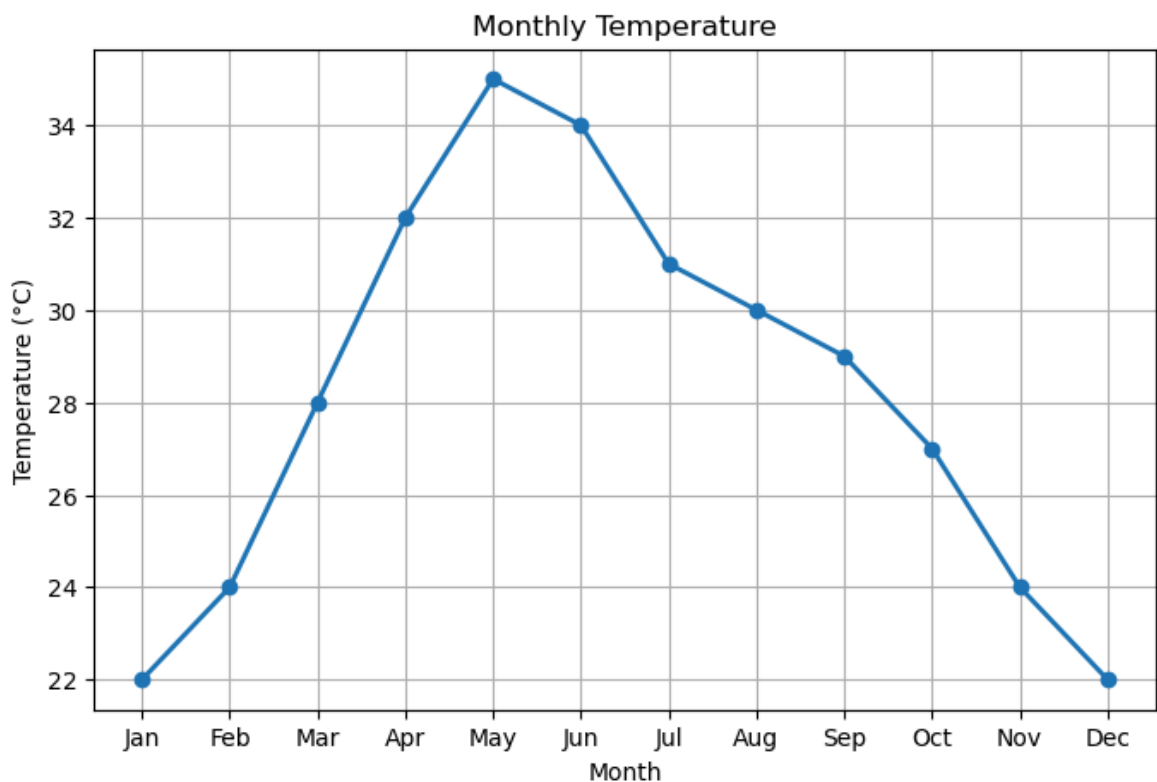
1. Create a Line Plot of Monthly Temperature

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

# Create DataFrame
data = {
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
              'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'],
    'Temperature': [22, 24, 28, 32, 35, 34, 31, 30, 29, 27, 24, 22],
    'Rainfall': [15, 20, 35, 50, 90, 180, 250, 230, 170, 100, 40, 20]
}

df = pd.DataFrame(data)

# Line Plot
plt.figure(figsize=(8,5))
plt.plot(df['Month'], df['Temperature'], marker='o', linewidth=2)
plt.title("Monthly Temperature")
plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
plt.grid(True)
plt.show()
```



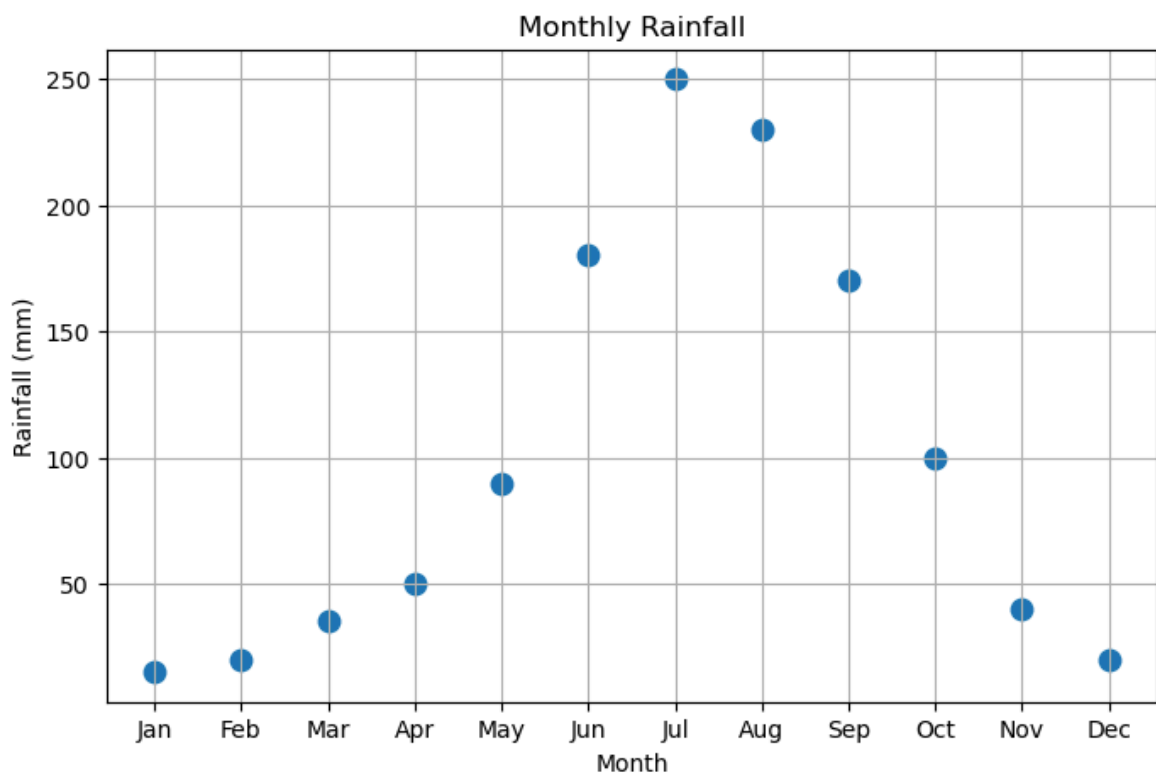
2. Create a Scatter Plot of Monthly Rainfall

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt

# Create DataFrame
data = {
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
              'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'],
    'Rainfall': [15, 20, 35, 50, 90, 180, 250, 230, 170, 100, 40, 20]
}

df = pd.DataFrame(data)

# Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(df['Month'], df['Rainfall'], s=80)
plt.title("Monthly Rainfall")
plt.xlabel("Month")
plt.ylabel("Rainfall (mm)")
plt.grid(True)
plt.show()
```



Q13. Scenario: Word Frequency Distribution from a Text File

The scenario clearly states read a text document named sample_text.txt.

sample_text.txt

Python is easy to learn.

Python is widely used in Data Science.

Data Science uses Python for data analysis, machine learning, and visualization.

Python helps in automation and scientific computing.

```
In [3]: from collections import Counter
import string

# Open and read the text file
with open("sample_text.txt", "r") as file:
    text = file.read().lower()

# Remove punctuation
text = text.translate(str.maketrans('', '', string.punctuation))

# Split text into words
words = text.split()

# Calculate frequency
frequency = Counter(words)

# Display frequency distribution
print("Word Frequency Distribution")

for word, count in frequency.items():
    print(word, ":", count)
```

Word Frequency Distribution

```
python : 4
is : 2
easy : 1
to : 1
learn : 1
widely : 1
used : 1
in : 2
data : 3
science : 2
uses : 1
for : 1
analysis : 1
machine : 1
learning : 1
and : 2
visualization : 1
helps : 1
automation : 1
scientific : 1
computing : 1
```

Q14. Scenario: Frequency Distribution of Customer Ages Using Pandas DataFrame

```
In [4]: import pandas as pd

# Sales Data
```

```

data = {
    'Customer_ID': [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    'Age': [22, 25, 22, 30, 28, 25, 22, 30, 35, 28]
}

# Create DataFrame
df = pd.DataFrame(data)

# Frequency Distribution
age_frequency = df['Age'].value_counts().sort_index()

print("Frequency Distribution of Customer Ages")
print(age_frequency)

```

Frequency Distribution of Customer Ages

Age

22 3

25 2

28 2

30 2

35 1

Name: count, dtype: int64

Q15. Scenario: Frequency Distribution of Likes on Social Media Posts

```

In [5]: import pandas as pd

# Social Media Dataset
data = {
    'Post_ID': ['P1', 'P2', 'P3', 'P4', 'P5', 'P6', 'P7', 'P8'],
    'Likes': [120, 200, 120, 300, 250, 200, 120, 300]
}

# Create DataFrame
df = pd.DataFrame(data)

# Frequency Distribution
likes_frequency = df['Likes'].value_counts().sort_index()

print("Frequency Distribution of Likes")
print(likes_frequency)

```

Frequency Distribution of Likes

Likes

120 3

200 2

250 1

300 2

Name: count, dtype: int64

In []: